

Supplementary Information

A Ground–Low-Altitude Mobility Manager for Service Reconfiguration under Disruptions

S1 Detailed experimental outcomes

The tables report service outcomes from the revised numerical analysis. Means use the original fixed scenario weights, and intervals use seed blocks. The B0 D60 results include the closed-loop duplicate-command correction. The version archive retains the original data and earlier tables.

Table S1: E1 service outcomes: mean [95% interval].

Site	Policy	n	Completion (s)	Deadline violations (%)	Damaged services/run
A	B0	240	187.67 [187.67, 187.67]	33.33 [33.33, 33.33]	0.000 [0.000, 0.000]
A	B1	240	160.78 [159.13, 162.42]	25.00 [25.00, 25.00]	0.500 [0.500, 0.500]
A	B2	240	147.62 [147.20, 148.04]	16.67 [16.67, 16.67]	0.058 [0.040, 0.077]
A	B4b	240	146.63 [145.88, 147.38]	16.67 [16.67, 16.67]	0.133 [0.114, 0.153]
B	B0	240	262.67 [262.67, 262.67]	50.00 [50.00, 50.00]	0.000 [0.000, 0.000]
B	B1	240	47.70 [46.49, 48.91]	0.00 [0.00, 0.00]	0.500 [0.500, 0.500]
B	B2	240	47.70 [46.49, 48.91]	0.00 [0.00, 0.00]	0.500 [0.500, 0.500]
B	B4b	240	49.10 [47.03, 51.17]	0.00 [0.00, 0.00]	0.487 [0.473, 0.502]
C	B0	240	323.67 [323.67, 323.67]	83.33 [83.33, 83.33]	0.000 [0.000, 0.000]
C	B1	240	168.28 [165.32, 171.23]	10.00 [4.12, 15.88]	0.500 [0.500, 0.500]
C	B2	240	168.28 [165.32, 171.23]	10.00 [4.12, 15.88]	0.500 [0.500, 0.500]
C	B4b	240	168.65 [165.58, 171.72]	10.00 [4.12, 15.88]	0.500 [0.500, 0.500]

Completion, damage and deadline-rate intervals use 20 seed blocks, each averaging the same 12 scenarios. A zero-width interval means that this fixed-panel average did not vary across seeds; it does not establish a universal rate. Each site–policy cell contains 240 runs.

Table S2: E2 all-run transport outcomes and conditional path restoration.

Site	Policy	All runs	Completion (s), mean [95% CI]	Deadline violations (%)	Recovered/affected	Restoration (%)
A	B0	320	182.00 [182.00, 182.00]	100.00	N/A	N/A
A	B1	320	162.00 [162.00, 162.00]	37.50	200/200	100.00
A	B2	320	162.00 [162.00, 162.00]	37.50	200/200	100.00
A	B4b	320	166.53 [164.41, 168.64]	39.06	191/191	100.00
B	B0	320	241.00 [241.00, 241.00]	100.00	N/A	N/A
B	B1	320	153.88 [153.88, 153.88]	37.50	200/200	100.00
B	B2	320	153.88 [153.88, 153.88]	37.50	200/200	100.00
B	B4b	320	154.05 [153.94, 154.16]	37.50	199/199	100.00
C	B0	320	310.00 [310.00, 310.00]	100.00	N/A	N/A
C	B1	320	278.00 [278.00, 278.00]	62.50	200/200	100.00
C	B2	320	278.00 [278.00, 278.00]	62.50	200/200	100.00
C	B4b	320	278.92 [277.79, 280.06]	62.81	201/201	100.00

All-run outcomes use 320 runs per site and policy. Recovery is conditional on the primary air chain actually being affected; it is N/A for B0. Restoration is reported as observed recovered/affected counts. We retain dependence within the fixed panel and do not apply a Wilson interval that assumes independent runs.

Table S3: Mean E3 system weighted loss by disturbance/task level.

Site	Level	Runs/policy	B0	B1	B2	B4b
A	L1	40	240.000	0.000	0.000	6.000
A	L2	80	240.000	240.000	240.000	240.000
A	L3	80	240.000	0.000	0.000	1.688
A	L4	40	240.000	120.000	120.000	123.375
B	L1	40	240.000	0.000	0.000	0.000
B	L2	80	240.000	120.000	120.000	120.000
B	L3	80	375.000	135.000	135.000	135.000
B	L4	40	375.000	255.000	255.000	255.000
C	L1	40	240.000	240.000	240.000	240.000
C	L2	80	240.000	240.000	240.000	240.000
C	L3	80	375.000	135.000	135.000	135.000
C	L4	40	375.000	375.000	375.000	375.000

L1/L4 have 40 runs and L2/L3 have 80 per policy. Levels change disturbance structure and demand. SWL uses flat configured penalties and priority weights, rather than monetary costs.

Table S4: E4A observation scheduling, completion and B4b input demand.

Interval (s)	B0 completion (s)	B1 completion (s)	B2 completion (s)	B4b completion (s)	B4b late (%)	B4b deci- sions/run	B4b prompt tokens/run
10	192.0	140.0	140.0	140.0	0	62	172849.0
30	212.0	150.0	150.0	150.0	0	22	62307.6
60	253.0	180.0	180.0	180.0	0	12	35017.4
120	253.0	240.0	240.0	240.0	100	7	21348.2
300	482.0	521.0	482.0	482.0	100	4	11707.6

Each policy–arm cell contains 20 runs. Decision counts cover the complete scheduled polling horizon, including post-resolution observations. Token means retain recorded backend outcomes.

Table S5: E4B injected execution delay and mean completion duration.

Delay (s)	B0 (s)	B1 (s)	B2 (s)	B4b (s)	B4b late (%)
0	182.0	120.0	120.0	120.0	0
1	183.0	121.0	121.0	121.0	0
5	187.0	125.0	125.0	125.0	0
10	192.0	130.0	130.0	130.0	0
20	202.0	140.0	140.0	140.0	0
30	212.0	150.0	150.0	150.0	0
60	242.0	341.0	302.0	302.0	100

Each policy–arm cell contains 20 runs. Waiting times are controlled simulation interventions, not measured inference latency.

Table S6: E4C B4b input burden with an unchanged core fleet.

Total aircraft records	Core aircraft	Completion (s)	On time (%)	Logged calls (20 runs)	Prompt tokens/logged call
	5	4 121.5	100	41	3982.1
	10	4 120.0	100	40	5920.6
	20	4 120.0	100	40	9588.7
	30	4 120.0	100	40	13256.7
	50	4 120.0	100	40	20593.5

Each arm contains 20 B4b runs. Added records represent unavailable and non-commandable aircraft. Token means include retained error calls with zero recorded tokens; active fleet size stays four.

Table S7: Aggregate E3 outcomes across the 12-scenario panel.

Site	Policy	n	Completion (s), mean [95% CI]	Deadline violations (%)	SWL, mean [95% CI]
A	B0	240	182.00 [182.00, 182.00]	100.00	240.000 [240.000, 240.000]
A	B1	240	187.25 [187.25, 187.25]	41.67	100.000 [100.000, 100.000]
A	B2	240	187.25 [187.25, 187.25]	41.67	100.000 [100.000, 100.000]
A	B4b	240	188.80 [187.18, 190.41]	42.08	102.125 [99.578, 104.672]
B	B0	240	241.00 [241.00, 241.00]	100.00	307.500 [307.500, 307.500]
B	B1	240	146.83 [146.83, 146.83]	25.00	127.500 [127.500, 127.500]
B	B2	240	146.83 [146.83, 146.83]	25.00	127.500 [127.500, 127.500]
B	B4b	240	147.05 [146.78, 147.32]	25.00	127.500 [127.500, 127.500]
C	B0	240	310.00 [310.00, 310.00]	100.00	307.500 [307.500, 307.500]
C	B1	240	322.67 [322.67, 322.67]	66.67	227.500 [227.500, 227.500]
C	B2	240	322.67 [322.67, 322.67]	66.67	227.500 [227.500, 227.500]
C	B4b	240	323.82 [322.97, 324.68]	66.67	227.500 [227.500, 227.500]

Scenario-seed runs receive equal weight. L2/L3 therefore contribute twice as many runs as L1/L4; equal-level weighting would define a different summary.

S2 Interface and implementation details

S2.1 Principal notation

Table S8: Principal notation.

Symbol	Meaning	Unit
$t, \tau(t), t_e, H$	Decision time, observation timestamp, scheduled execution and horizon	s
$S_t, \hat{S}_t$	Current joint state and observed state	—
r_m, d_m, c_m	Mission release, deadline and completion	s
$\mathcal{I}_t, \mathcal{C}_{m,t}$	Candidate information and legal subset	—
$\hat{\eta}_c, \delta_{\text{exec}}, \sigma_m$	Remaining completion estimate, execution delay, estimated temporal margin	s
$a^{\text{prop}}, u, Q, \Gamma$	Proposal, executed intervention, current-version indicator, current-state checker	—
$J_t(c)$	Specified one-step heuristic score	Configured seconds-equivalent units
A_i, R_i	Affected-chain and recovered-path indicators	Binary
T_m, V_m, L_H	Completion duration, deadline violation, weighted service loss	s; binary; loss units

S2.2 Structured action vocabulary

Table S9: Structured action vocabulary.

Action	Meaning
DISPATCH	Assign an available compatible aircraft
REASSIGN	Transfer an eligible occupied aircraft to another mission
REROUTE	Change an ongoing operation’s route
DIVERT	Change an air operation’s destination or continuation
DELAY	Postpone an operation
CANCEL	Cancel a specified operation
RESERVE	Reserve a resource for a mission
RETURN	Request return by an eligible aircraft
LAND	Request landing under modeled constraints
GROUND_FALLBACK	Establish ground service for an eligible mission
NO_ACTION	Make no operational change at the decision epoch
ESCALATE	Report a need beyond the current action choice

Each scenario uses only part of this vocabulary. Including an action type in the contract does not mean that the experiments validate every possible use of it. The candidate interface records legal flags and rejection reasons, and the checker applies conditions specific to each action.

S2.3 Policy inputs and parameters

The complete template is supplied as `reproducibility/prompts/manager_v2.txt`, together with the input renderer and action schema. Both B4a and B4b receive the full global snapshot without its `candidates` field. B4b receives that field in a separate JSON section; B4a does not. Both retain raw resource state, mission assignments, priorities, positions, deadlines, ground ETA, prompt rules and the common checker. Only B4b receives derived air ETA, completion predictions, per-candidate legality/rejection reasons and the explicit mapping to preempted missions. The ablation therefore tests explicit candidate-information support, including both derived information and its presentation.

Table S10: Input-field comparison for the E1 ablation.

Field	B4a	B4b
Raw aircraft and mission state; ground ETA	Included	Included
Rules, objectives, output schema, checker	Same	Same
Derived candidate air ETA and completion time	Omitted	Included
Per-candidate legal flag and rejection reason	Omitted	Included
Predicted deadline violation and preemption consequence	Omitted	Included

Table S11: Key operational and analysis parameters.

Parameter	Value or implemented rule
Core fleet	Two medical UAVs, one logistics UAV, one passenger eVTOL
Cruise speed	Medical/logistics 15 m/s; passenger 40 m/s; inspection role 10 m/s (no core inspection aircraft)
Nominal altitude	UAV 328 ft; passenger eVTOL 656 ft
Arrival radius	100 m; this can make realized duration shorter than geometric ETA
Dispatch overhead	0 s in candidate estimates
Endurance	Initial 1,800 s, reduced by a seeded 0–15%; candidate reserve margin 60 s
Busy-aircraft progress	Seeded fraction 0.05–0.40 of its initial route
Simulation step / ordinary decision period	1 s / 30 s while actionable; E4A overrides observation polling
Horizon / primary emergency release	900 s (E3 L3/L4: 1,200 s) / 300 s
E1 deadline slack	HIGH 300 s; CRITICAL 180 s
E2/E4 primary deadline	480 s at Site A; site-adapted configurations accompany the data
Incumbent default deadline	1,800 s for initial background missions
Ground completion	Execution-time SUMO D1–H1 ETA rounded to seconds, added to issue time
LLM decoding	Temperature 0; JSON output; 8,192 output-token budget
SWL priority weights	CRITICAL/HIGH/NORMAL/LOW: 4/3/2/1; penalties are task-specific flat values

B2 weights and the prompt template remain unchanged. The supplied YAML matrices specify mission-specific delay and cancellation penalties, secondary releases, workload assignments and site adaptations. The policy identifier names the configured backend service; the underlying model version has not been verified.

S3 Statistical and outcome conventions

S3.1 Fixed-panel estimand and random streams

The runner initializes a fresh `random.Random(seed)` for every scenario and visits aircraft in sorted order. Each aircraft consumes an endurance draw, and each busy aircraft consumes an additional progress draw. SUMO also reuses the same seed. Workload-dependent draw counts change the realizations, but they do not create independently keyed scenario streams. We therefore retain dependence within each seed across the fixed scenario panel and pair policies by scenario and seed. This decision follows the implementation, rather than seed labels or state hashes alone.

For a complete panel, each seed contributes the average of its scenario-specific outcomes or paired differences. Means keep equal scenario–run weights. A t interval uses the standard deviation across 20 seed averages (10 in the E1 ablation). For a conditional endpoint with N retained pairs, let $\bar{d}$ denote the original pair-weighted mean and K the number of represented seeds. The cluster contribution is

$$B_s = \bar{d} + \frac{K}{N} \sum_{i: s_i=s} (d_i - \bar{d}).$$

The interval is $\bar{d} \pm t_{.975, K-1} \text{sd}(B_s) / \sqrt{K}$. This retains the mean over jointly affected pairs without reweighting seeds that have different exposure counts. Conditional populations can differ by policy. At Site C, B4b has 201 affected runs, whereas the B4b–B2 recovery comparison includes 200 jointly affected pairs.

Continuous tests use these seed contributions, with the original signed-rank fallback for constant differences. Binary tests enumerate policy-label swaps of whole seed blocks and use the total discordant paired difference as the test statistic. This preserves pairing without assuming independent pairs for McNemar inference. Scenarios, endpoints and correction families are not selected by their p-values. Intervals are unadjusted. Holm-adjusted p-values retain the original three E1 ablation, 12 E2 and 12 E3

compound endpoints per site. E1 scenario-specific and E4 single-scenario comparisons keep their original seed units.

The E2 family includes completion, recovery time, replan latency, incumbent damage, candidate reduction, deadline violation, restoration, recovered completion, ground fallback, failure-induced fallback, necessary-fallback correctness and air intervention. E3 compares B4b with B1/B2 at L2–L4 for SWL and CRITICAL+HIGH loss.

Table S12: Complete E2 seed-block comparison family, Site A. Differences are B4b minus B2; binary differences are fractions.

Site	Endpoint	Pairs	Difference	95% interval	Holm p
A	Completion	320	4.525	[2.412, 6.638]	0.003069
A	Recovery time	191	3.927	[0.930, 6.924]	0.142465
A	Replan latency	191	3.927	[0.930, 6.924]	0.142465
A	Damage	320	0.003	[-0.012, 0.018]	1.000
A	Candidate reduction	320	0.013	[0.000, 0.025]	0.336690
A	Deadline risk	320	0.016	[0.003, 0.029]	0.437500
A	Restoration	191	0.000	[0.000, 0.000]	1.000
A	Recovered/completed	191	0.000	[0.000, 0.000]	1.000
A	Ground fallback	320	0.013	[0.000, 0.025]	0.625000
A	Failure fallback	320	0.013	[0.000, 0.025]	0.625000
A	Necessary fallback	320	-0.016	[-0.029, -0.003]	0.437500
A	Air intervention	320	-0.028	[-0.048, -0.008]	0.142465

Table S13: Complete E2 seed-block comparison family, Site B. Differences are B4b minus B2; binary differences are fractions.

Site	Endpoint	Pairs	Difference	95% interval	Holm p
B	Completion	320	0.175	[0.060, 0.290]	0.055710
B	Recovery time	199	0.281	[0.098, 0.465]	0.055710
B	Replan latency	199	0.281	[0.098, 0.465]	0.055710
B	Damage	320	0.000	[0.000, 0.000]	1.000
B	Candidate reduction	320	0.000	[0.000, 0.000]	1.000
B	Deadline risk	320	0.000	[0.000, 0.000]	1.000
B	Restoration	199	0.000	[0.000, 0.000]	1.000
B	Recovered/completed	199	0.000	[0.000, 0.000]	1.000
B	Ground fallback	320	0.000	[0.000, 0.000]	1.000
B	Failure fallback	320	0.000	[0.000, 0.000]	1.000
B	Necessary fallback	320	-0.003	[-0.010, 0.003]	1.000
B	Air intervention	320	-0.003	[-0.010, 0.003]	1.000

Table S14: Complete E2 seed-block comparison family, Site C. Differences are B4b minus B2; binary differences are fractions.

Site	Endpoint	Pairs	Difference	95% interval	Holm p
C	Completion	320	0.922	[-0.214, 2.057]	1.000
C	Recovery time	200	0.255	[-0.138, 0.648]	1.000
C	Replan latency	200	0.255	[-0.138, 0.648]	1.000
C	Damage	320	0.000	[0.000, 0.000]	1.000
C	Candidate reduction	320	0.000	[0.000, 0.000]	1.000
C	Deadline risk	320	0.003	[-0.003, 0.010]	1.000
C	Restoration	200	0.000	[0.000, 0.000]	1.000
C	Recovered/completed	200	0.000	[0.000, 0.000]	1.000
C	Ground fallback	320	0.000	[0.000, 0.000]	1.000
C	Failure fallback	320	0.000	[0.000, 0.000]	1.000
C	Necessary fallback	320	0.000	[0.000, 0.000]	1.000
C	Air intervention	320	0.000	[0.000, 0.000]	1.000

Table S15: E3 compound SWL comparisons. CRITICAL+HIGH differences and tests are identical in these cells; both endpoints remain in each 12-test family.

Site	Level	Base	Pairs	Difference	95% interval	Holm p
A	L2	B1	80	0.000	[0.000, 0.000]	1.000
A	L3	B1	80	1.688	[-1.844, 5.219]	1.000
A	L4	B1	40	3.375	[-3.689, 10.439]	1.000
A	L2	B2	80	0.000	[0.000, 0.000]	1.000
A	L3	B2	80	1.688	[-1.844, 5.219]	1.000
A	L4	B2	40	3.375	[-3.689, 10.439]	1.000
B	L2	B1	80	0.000	[0.000, 0.000]	1.000
B	L3	B1	80	0.000	[0.000, 0.000]	1.000
B	L4	B1	40	0.000	[0.000, 0.000]	1.000
B	L2	B2	80	0.000	[0.000, 0.000]	1.000
B	L3	B2	80	0.000	[0.000, 0.000]	1.000
B	L4	B2	40	0.000	[0.000, 0.000]	1.000
C	L2	B1	80	0.000	[0.000, 0.000]	1.000
C	L3	B1	80	0.000	[0.000, 0.000]	1.000
C	L4	B1	40	0.000	[0.000, 0.000]	1.000
C	L2	B2	80	0.000	[0.000, 0.000]	1.000
C	L3	B2	80	0.000	[0.000, 0.000]	1.000
C	L4	B2	40	0.000	[0.000, 0.000]	1.000

S3.2 Service-loss audit

All 2,880 E3 run registries were checked against their recorded horizons: 900 s for L1/L2 and 1,200 s for L3/L4. At each site, 1,440 task instances had deadlines at or before the corresponding horizon, and all were COMPLETED. Across sites, zero overdue tasks remained EN_ROUTE or ASSIGNED. The audit therefore requires no correction to SWL for overdue ongoing tasks. Background missions with 1,800 s deadlines fall outside both horizons; remaining in service does not by itself constitute a failure. Incumbent damage remains a count of damaged final states.

As a complementary E1 measure, we linked recorded preemption times to the incumbent identified in the candidate log and its final interrupted state. At A, exposure from preemption to the horizon for persistently interrupted services averages 35 s per run for B2 and 80 s for B4b (B1: 300 s). This measure is truncated at the horizon; it does not capture the complete cumulative cost of interruption. Mission CSV records omit some incumbent preemption transitions, so a logged interruption duration of zero would be misleading. Initial incumbent deadlines are 1,800 s, beyond the 900 s horizon. This exposure measure therefore supports no claim about incumbent lateness.

S4 Technical checks and timing sensitivity

S4.1 Duplicate pending commands and E4B regression

The original B0 D60 pipeline records a GROUND_FALLBACK command at 300 s, due at 360 s. At 360 s, the fault triggers a new decision with the same normalized action and the same 181.763 s ETA field. The original queue replaces the pending command and restarts the 60 s wait. It issues the command at 420 s, and transport completes at 602 s. This extra wait produced the original 302 s completion duration.

The revised queue keeps the original due time when a validated normalized action exactly matches a pending command. A changed action still supersedes the pending command, and the surviving command is revalidated at execution. The `queue_fix.py` overlay implements this correction without changing the original project source or raw runs.

All 20 B0 D60 seeds were rerun through SUMO–BlueSky. The command issued at 360 s, and transport completed at 542 s, giving a duration of 242 s. Single-seed D00/D30 controls retain 182/212 s; fault times 359/361 s both give 242 s. B2’s changed air-to-ground command retains a 302 s duration. B1 retains its air-to-air replacement sequence. Patched and unpatched control reruns both give 342 s

in the current runtime, compared with the archived 341 s. This 1 s reproduction difference is unrelated to deduplication. We retain the archived cell because the correction does not change the command sequence. These regressions made no LLM calls. The original B4b traces contain genuine command replacement and remain unchanged.

Table S5, main Figure 6 and Supplementary Figure S2 update only the affected B0 cell. D60 tests a specified event order; it does not establish a universal transport-delay threshold. Backend wall-clock time remains separate from controlled simulation waiting.

S4.2 Physical transfer semantics and offline added-time sensitivity

The model replaces service assignments without tracking individual payloads. New air candidates route through the mission origin. Interrupted-service candidates route through V3 before the destination, but the supervisor can dispatch a backup before the original carrier reaches V3. Preemption interrupts and detaches the incumbent mission. Ground takeover starts a service timer based on the current reference corridor ETA. The model has no explicit states for unloading, rendezvous, loading, new inventory or re-shipment.

We evaluate added transfer time through an *offline trajectory-fixed sensitivity*, rather than a new closed-loop experiment. For an actually affected E2 mission with a post-failure DISPATCH, REASSIGN or GROUND_FALLBACK, we add one hypothesized transfer delay h to recorded completion. The rule is the same for every policy; uninterrupted services receive no added time. A previously timely completion becomes late when $0 \leq d_m - c_m < h$. Choices, fault exposure and later scheduling remain fixed. The values $h = 15, 30, 60$ s are illustrative assumptions, not measured handling durations.

Table S16: Offline added transfer time: number of previously timely E2 replacements becoming late.

Site	Policy	Replacements	Timely	+15 s	+30 s	+60 s
A	B1	200	80	0	0	0
A	B2	200	80	0	0	0
A	B4b	191	75	1	1	2
B	B1	200	80	0	80	80
B	B2	200	80	0	80	80
B	B4b	199	80	0	80	80
C	B1	200	0	0	0	0
C	B2	200	0	0	0	0
C	B4b	201	0	0	0	0

At Site B, 80 affected replacements per coordinated policy are timely with 19 s remaining; adding 30 s makes all 80 late. At A, timely B1/B2 replacements have 60 s remaining, and none becomes late for $h \leq 60$ s. B4b has one such change at 15/30 s and two at 60 s. At C, the affected replacements in this subset are already late.

In the preemption example in Section 4.2, an extra 17 s of air-specific handling erases the realized speed advantage of 212 versus 229 s. More than 88 s also causes the emergency to miss its deadline. Adding the same handling time to both initial alternatives preserves their speed difference, so this threshold concerns an additional preemption step specifically. These checks identify conclusions that depend on omitted stages. They do not validate dynamic rescheduling with payload transfer.

S4.3 Road-statistics boundary

The original morphology routine summed all imported passenger edges and divided by 10.24 km^2 , without clipping them to its bbox argument. The revised calculation intersects each edge polyline with the configured geographic rectangle after projection through the SUMO network transform. It apportions SUMO edge lengths by the fraction of the polyline inside the rectangle and uses the same polygon’s area for density. Intersection counts include only nodes inside the polygon, but count their distinct neighbors in the full graph to avoid artificial boundary intersections. Original full-network topology, OD samples and routing remain separately identified and unchanged. Site C still has higher intersection density than A; its description instead draws on dead ends, degree, circuitry and facility distance.

S4.4 Fault state machines and recovery conditions

Table S17: Six modeled fault families. Exact targets, geometry, times and windows are supplied in the YAML matrices.

Fault	Trigger and state change	Independent local response	Service-recovery condition
F1 C2 loss	Target becomes non-commandable and in CONTINGENCY; mission interrupted	Return to V3	Surviving compatible backup or ground; lost target remains excluded
F2 GNSS degradation	One-shot test of aircraft inside configured zone; GNSS DEGRADED	Occupied affected aircraft abandon mission and divert to V3	Alternative with acceptable navigation and landing state; zone alone does not block other routes
F3 UTM change	DEGRADED blocks new noncritical air assignments; OUTAGE blocks all new air	OUTAGE interrupts in-flight missions and diverts them to V3	Ground during outage; otherwise an eligible air assignment
F4 Landing closure	Configured site becomes unavailable; missions to it interrupted	Affected aircraft divert to V3	Available compatible destination/continuation or ground
F5 Unknown aircraft	Prescribed noncooperative trajectory and temporary risk zone; affected missions interrupted	Commandable affected aircraft divert to V3	Aircraft and route outside active zone, or ground; zone expires at configured end
F6 Flyaway	Target becomes uncontrolled and non-commandable; surrounding envelope can interrupt others	Target follows prescribed uncontrolled trajectory; other commandable affected aircraft divert	Exclude target and active envelope; use surviving air or ground

The decision and execution checks evaluate whether zones apply within their configured active windows. Communication, navigation, UTM and landing-site states are not automatically repaired during these runs. Recovery means establishing a replacement service, not repairing the original aircraft or infrastructure.

S5 Figure production and source records

Figure 1 combines separate, text-free AI-generated illustrations with editable PowerPoint text, routes, containers and markers. The map is labeled Not to scale; the finding graphics are schematic. The revision retains the authors’ later PowerPoint edits removing the duplicate figure title and adjusting module labels. The authors supplied the visual reference. The illustrations depict no observations or operational deployments at the study sites. They were produced with the built-in imagegen tool and composed in PowerPoint. The records identify the tool without assigning an unverifiable model-release version.

All figure text uses Times New Roman. Python generates editable SVG/PDF vectors for main Figures 2–6 and S3 from unchanged geometry and result summaries. Figure 5’s paired contrasts and original intervals appear in S3. Figure 1 retains its PowerPoint source; S1 retains charts with embedded workbooks, and S2 editable command and transport segments. Previous documents and decks are archived. The editing guide and `FIGURE_STYLE_GUIDE.md` record current sources and design rules.

S6 Reproduction files and entry points

No public repository URL has been assigned. `REPRODUCTION_README.md` and `reproduce_analysis.ps1` specify original run locations, dependencies and executable processing order. The `reproducibility/` folder contains prompts, schemas, weights, experiment configurations, derived data and a hashed file inventory. The package includes the local correction overlay, an optional regression entry point that makes no LLM calls, editable figure builders and PowerPoint sources.

S7 Supporting transport outcomes and command sequence

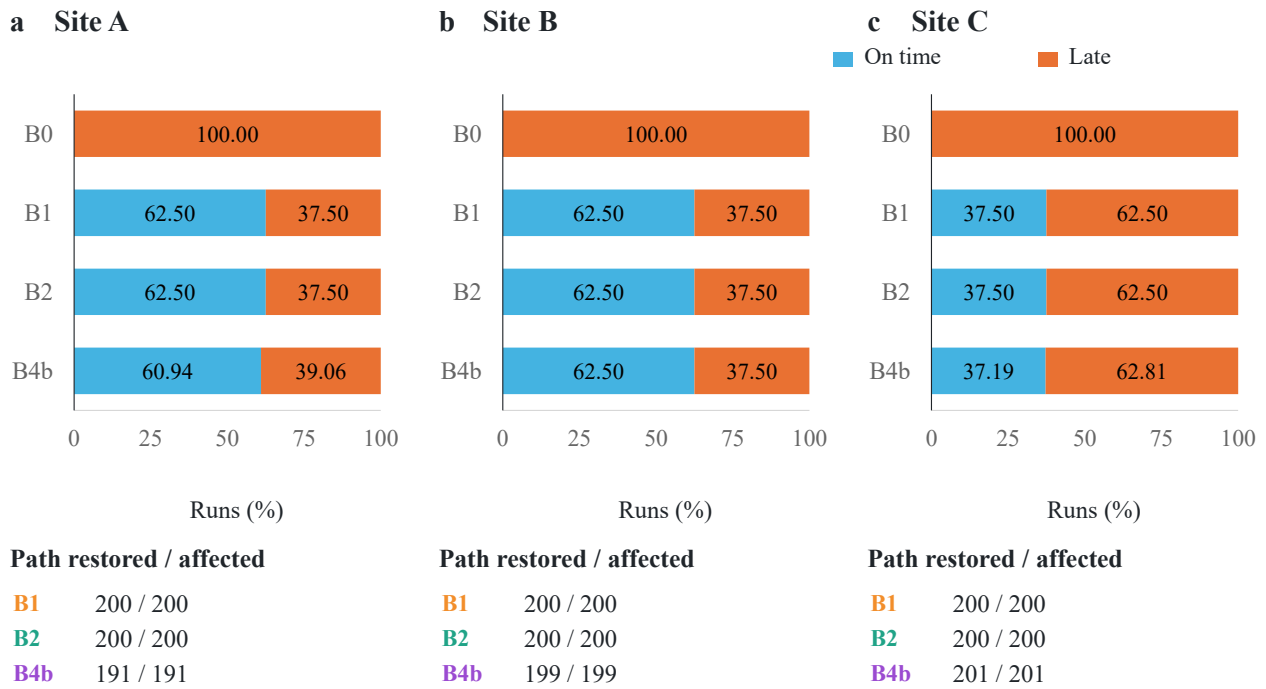


Figure S1: E2 path restoration and timely service answer different questions. (a–c) Bars show on-time and late completion percentages over all 320 runs per site and policy. The separate counts below give restored/actually affected primary air chains for B1/B2/B4b. Recovery denominators can depend on prior policy actions; B0 has no affected-air-chain denominator. Blue and orange encode deadline outcome here, not policy identity. Table S2 retains the full summaries and exact restoration counts.

a B0: repeated ground command



b B4b: air replaced by ground

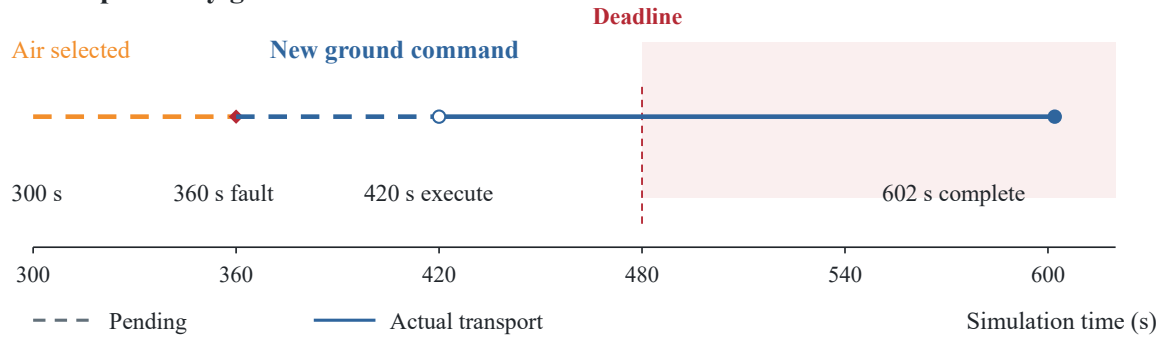


Figure S2: E4B D60 command sequences after identical-command deduplication. The pending air dispatch is superseded by a ground-fallback decision at the fault time. Ground executes at 420 s and completes at 602 s, after the 480 s deadline. Orange denotes the air command, blue the ground service and red the fault/deadline. Dashed lines denote pending commands and solid lines actual transport. Corrected B0 ground service executes at 360 s and completes at 542 s; B4b changes mode and completes at 602 s.



Paired means and 95% seed-block intervals · Negative values indicate lower loss under B4b
 40 runs in L1/L4; 80 in L2/L3 per policy · Levels jointly vary disturbance and demand

Figure S3: E3 paired coordination benefit by site and compound level. Points show mean B4b-minus-B0 differences in system weighted loss; whiskers are the original 95% seed-block intervals. Negative values indicate lower loss under B4b. Zero-width intervals coincide with their points. Each policy has 40 runs in L1/L4 and 80 in L2/L3. Levels jointly vary disturbance and demand. The corresponding policy-level means appear in main Figure 5; exact values remain in Tables S3 and S7.